

Kubernetes: Breathing New Life into On-Premise Services



DevOps is to IT what IoT is to manufacturing. Both are disrupting the *status quo* and are taking the world closer to automation. Let us look at one of the biggest drivers of this revolution — Kubernetes.

Containers have made a visible impact on software development over the last six years. They have the ability to co-locate, encapsulate and isolate application environments, yet abstract the complexity of the underlying OS and physical hardware. This results in a transformative shift in data centre operations — from a machine-centric one to an application-centric one. However, the manual efforts required to manage a group of containers inhibit their large scale deployment and proliferation. Docker Swarm attempted to solve this problem but lost to the Goliath, i.e., Kubernetes, which

innumerable users will vouch for. Docker Inc. itself has now included Kubernetes in its distribution alongside Docker Swarm, which endorses Kubernetes capabilities. This should not surprise anybody.

Though Kubernetes was officially launched in 2014, its predecessor, Borg, has a reputation for running millions of jobs in isolated application-led environments in Google's data centres, which host all its offerings. As containers gained ground, Google ported Borg from C++ to Go, applied lessons learned and re-christened it Kubernetes. It then donated Kubernetes to the Cloud Native Computing Foundation